Cardiac Deformation Lab

Image-Based Cardiac Deformation Mapping & Kinematic Strain Analysis

Analysis ID:	f8465126-d147-4133-9390-a86e89219ad5	**Analysis Type:**	Single-Pair Differential Strain Analysis
Execution Date:	October 16, 2026 at 05:02:41 UTC	**Motion Model:**	Dense Farnebäck Optical Flow
Spatial Calibration:	Not provided; displacement values reported in image pixels (px)	**Calibration Status:**	Image-Space Only
Analysis Region:	Full Image (All image-space pixels evaluated)	**Region Scope:**	Full Image
Reference Frame (ED):	qc_good_1.png	**Target Frame (ES):**	qc_good_2.png

Computational Analysis Quality

Evaluation of measurable computational properties (image quality, optical-flow stability, strain tensor validity, and region scope).

Quality Score:	**100 / 100**	**Quality Level:**	**High**

Quality Component	Technical Status	Component Score	Weight
Image Quality	Good	100 / 100	25.0%
Optical Flow Quality	Good	100 / 100	35.0%
Strain Tensor Consistency	Good	100 / 100	40.0%
Region of Interest (ROI)	Not Applicable	100 / 100	0.0%
ROI Tracking Reliability	Not Applicable	100 / 100	0.0%
Temporal Sequence Dynamics	Not Applicable	100 / 100	0.0%
Spatial Calibration Consistency	Uncalibrated	100 / 100	—

Informational Notices:
- Input frames possess identical dimensions (128x128).
- High motion coherence (0.88) indicates smooth flow alignment.
- Full-image analysis mode active; no localized ROI masking applied.
- Automated ROI tracking was not active for this analysis.
- Single-pair evaluation; multi-frame temporal continuity not applicable.
- Spatial calibration was not provided; displacement remains in image-space units.

Notice: This quality assessment describes computational properties of the image-analysis pipeline. It is not a clinical confidence score or diagnostic assessment.

1. Displacement & Kinematic Motion

Metric Parameter	Value (Pixel Units)	Description

Mean Displacement	0.8088 px	Average Euclidean magnitude of displacement vector field
Maximum Displacement	5.7985 px	Peak regional apparent motion amplitude across the field
Minimum Displacement	0.0000 px	Lowest regional motion magnitude observed
Displacement Std. Dev.	1.6450 px	Spatial heterogeneity of tissue displacement

2. Small-Deformation Strain Tensor & Principal Fields

Strain Component	Mean	Min	Max	Std Dev
Normal Exx (du/dx)	7.2831e-11	N/A	1.1937	N/A
Normal Eyy (dv/dy)	9.3132e-10	N/A	1.0676	N/A
Shear Exy (0.5*(du/dy+dv/dx))	4.6566e-10	N/A	1.1839	N/A
Principal Strain 1 (Major)	0.0433	N/A	N/A	N/A
Principal Strain 2 (Minor)	-0.0433	N/A	N/A	N/A

3. Advanced Deformation Metrics & Geometric Projection

Metric Parameter	Observed Value	Physical / Mathematical Interpretation
Motion Coherence	0.8761	Cosine consistency of motion vectors within local 5x5 spatial kernels
Strain Gradient Magnitude	0.0273	Spatial rate of change of principal strain fields
Strain Anisotropy Index	0.3974	Directional preference of tissue deformation (|e1 - e2| / (|e1| + |e2|))
Strain Entropy	2.0596	Shannon entropy describing dispersion of regional strain histograms
Deformation Energy	0.0451	Pseudo-elastic strain energy density surrogate (0.5 * Tr(eps * eps))
Radial Deformation (Mean)	0.3226	Motion projection along radial axes from ventricular centroid approximation
Circumferential Deformation (Mean)	0.0020	Motion projection orthogonal to radial vectors (tangential shortening)

5. Cardiac Deformation Fingerprint (10-Dimensional)

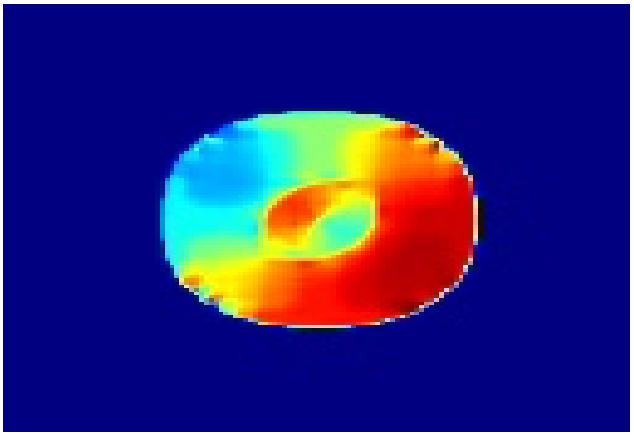
Dimension Symbol	Canonical Name	Value
D_mean	Mean Displacement	N/A
D_max	Maximum Displacement	N/A
epsilon_mean	Representative Mean Strain	N/A
epsilon_max	Representative Max Strain	N/A
epsilon_principal	Representative Principal Strain	N/A
AI	Strain Anisotropy Index	N/A

H	Strain Entropy	N/A
G_epsilon	Strain Gradient Magnitude	N/A
C	Motion Coherence	N/A
TSI	Temporal Stability Index	N/A

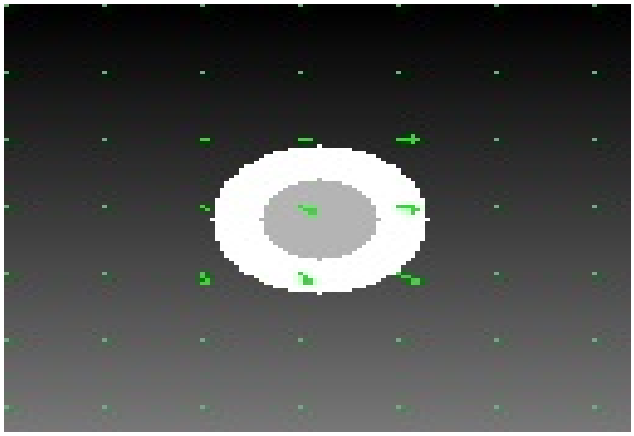
6. Spatial Deformation Maps & Motion Field Visualizations

Two-dimensional colormaps generated by OpenCV and matplotlib engines illustrating spatial gradients and vector trajectories.

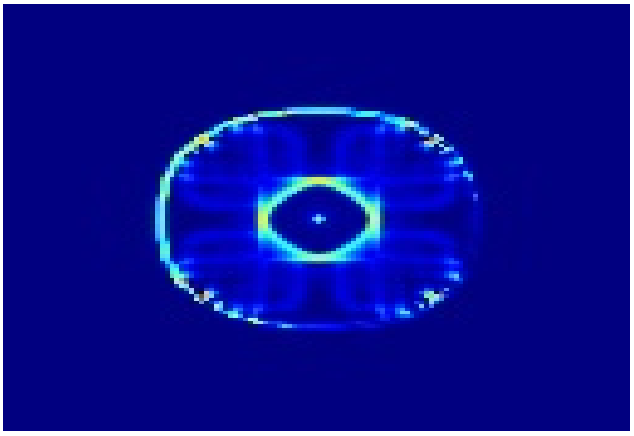
Displacement Heatmap



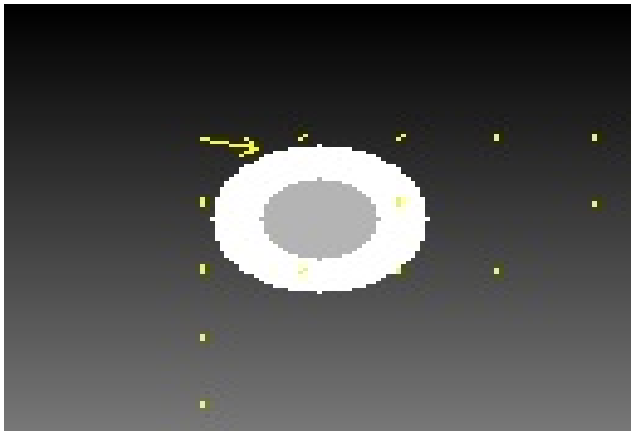
Motion Vector Map



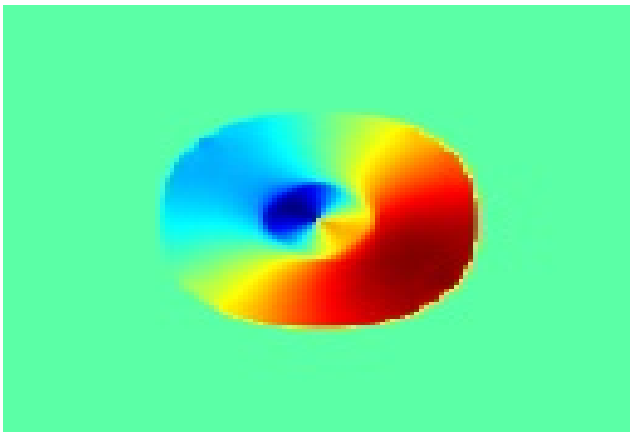
Principal Strain Heatmap



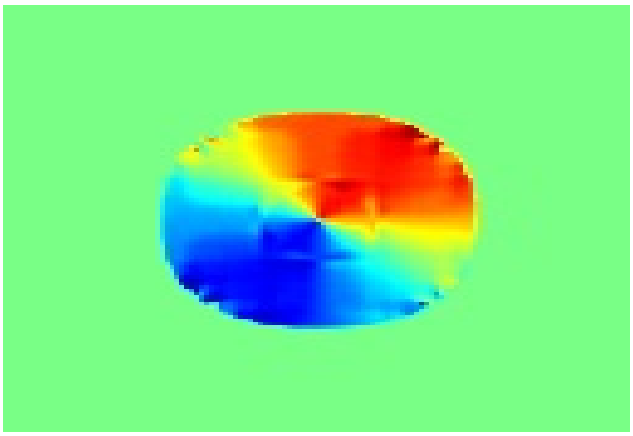
Principal Direction Map



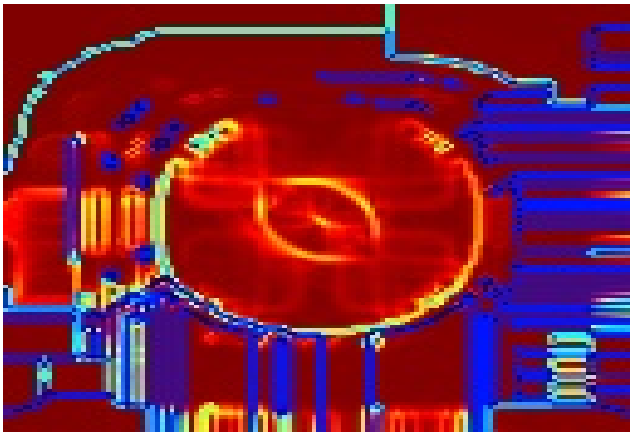
Radial Deformation Map



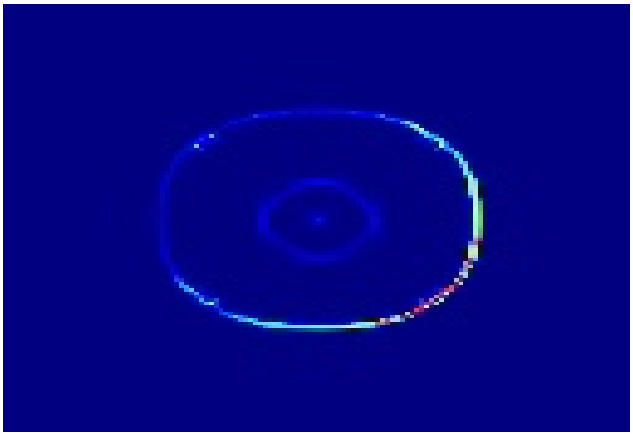
Circumferential Deformation



Motion Coherence Map



Deformation Energy Map



7. Technical Interpretation & Research Prototype Disclaimer

Methodological Scope: The metrics, vector fields, and spatial colormaps detailed in this report represent apparent image-space deformation computed via Farnebäck dense optical flow and classical continuum strain tensor decomposition. Measurements are expressed in image coordinate units (pixels) and relative mathematical differentials.

RESEARCH ONLY — NO CLINICAL DIAGNOSTIC ASSERTIONS:

This software is an experimental computational research prototype. All values, including the Cardiac Deformation Fingerprint (10-D) and Temporal Stability Index (TSI), are algorithmic descriptors of pixel kinematics. This system does NOT provide medical diagnoses, clinical risk assessments, normal/abnormal classifications, or treatment recommendations. This report must NOT be used as a medical device or diagnostic decision-making instrument.